Finding Nothing Reliably

A practical guide to systematic trading research without false positives, from probability basics to a verdict you can trust

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ABSTRACT

Backtests are cheap, and with enough of them anything looks profitable. Most strategies that look good in research fail in live trading. The usual causes are not exotic: they are sampling noise, many attempts, information from the future leaking into the test, and costs. This paper is a self-contained introduction to the discipline that guards against those failures, written for a reader who directs an AI agent through the research and needs to judge what the agent reports. We cover the probability you need (expectation, standard error, the Sharpe ratio and its uncertainty, base rates), the ways a backtest fools you (multiple testing, overfitting, leakage, costs), and a twelve-stage workflow from idea to deployment that is built so every failure is recorded and nothing is quietly re-run. We explain the tools that put a number on luck: the deflated Sharpe ratio, the probability of backtest overfitting, false discovery rate control, power analysis, placebos and sequential tests. Then we show how to read a verdict. Every figure is simulated from a stated seed, so the reader can see each failure happen on data that has no edge. The companion specification, `FRAMEWORK.md`, tells an agent how to build the workspace described here.

Keywords: backtest overfitting; multiple testing; deflated Sharpe ratio; data leakage; pre-registration; out-of-sample validation; prediction markets; research workflow.

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1 Introduction

A researcher who tests one hundred ideas on the same data, each with no edge at all, will usually find several that look excellent. Nothing went wrong with the arithmetic. That is what randomness does when you search it. Markets are competitive, transaction costs are certain, and real edges are small. A process that is not designed against false positives will mostly produce them.

The goal of a research framework is therefore odd at first sight. It is not to find strategies. It is to make believing a false one expensive, and to close a dead idea quickly, permanently and on the record. A good framework ends most ideas with "no edge detected, and here is how large an edge we could have seen". When something does survive, the survival means something because the bar was set before anyone looked.

This paper explains that discipline from the ground up. Section 2 covers the probability you need. Section 3 shows, on simulated data with no edge, the four main ways a backtest lies. Section 4 walks through the workflow from idea to deployment. Section 5 covers the statistical tools against luck and overfitting. Section 6 covers data. Section 7 covers intraday trading,

long-horizon portfolios and prediction markets. Section 8 is a field guide to reading results. Section 9 covers working with an AI agent. Formulas are collected in Appendix A and terms in Appendix B.

> THE ONE-SENTENCE VERSION
>
> Decide everything before you look, count every look, judge effect sizes rather than p-values, spend unseen data once, and treat "inconclusive" as an honest answer rather than a failure.

2 The probability you need

2.1 A trade is a random variable

The outcome of a trade is not known in advance. It is a *random variable*: a quantity with a distribution of possible values. Two numbers summarise most of what matters. The *expectation* (mean) μ is the average outcome over many repetitions. The *standard deviation* σ is the typical size of the deviation from that average. For a strategy that wins W with probability p and loses L otherwise,

$$\mathbb{E}[\text{P\&L}] = p\,W - (1-p)\,L.$$

A 55% win rate with equal wins and losses of 10 ticks has an expectancy of $0.55 \cdot 10 - 0.45 \cdot 10 = 1$ tick per trade, *before costs*. If a round trip costs 1.5 ticks, the expectancy is -0.5 ticks. A high win rate says nothing on its own. A strategy that wins 90% of the time and loses 20 times its average win when it loses has negative expectancy.

2.2 Sampling noise and the square-root law

We never observe μ. We observe an average $\bar{x}$ over n trades, and that average is itself random. Its standard deviation, the *standard error*, is

$$\mathrm{SE}(\bar{x}) = \frac{\sigma}{\sqrt{n}}.$$

This square-root law governs everything that follows. Quadrupling the sample only halves the noise. With a per-trade standard deviation of 12 ticks and 500 trades, the standard error is $12/\sqrt{500} \approx 0.54$ ticks, so a true edge of 1 tick sits less than two standard errors from zero. Real edges are usually small compared with the noise of each trade. That is why so many trades are needed before anything can be said.

2.3 Independence, clusters and fat tails

The formula above assumes independent observations. Trades on the same day share the same market conditions, so they are not independent. Twenty trades on one volatile afternoon are closer to one observation than to twenty. The honest unit is often the *cluster*: the session, the rebalance period or the event. Returns also have *fat tails* (extreme moves happen more often than a normal distribution predicts) and *volatility clustering* (calm and turbulent periods persist). Both make naive standard errors too small. The standard remedy is the *block bootstrap*: resample whole blocks of consecutive clusters, so the dependence inside a block is preserved, and read the uncertainty off the resampled distribution.

2.4 The Sharpe ratio and how uncertain it is

The Sharpe ratio is the mean return divided by its standard deviation, $\mathrm{SR} = \mu/\sigma$, usually annualised by multiplying by $\sqrt{252}$ for daily data. Its estimate is noisy. For roughly normal returns, the standard error of an annualised Sharpe estimated from T years is about

$$\mathrm{SE}(\widehat{\mathrm{SR}}) \approx \sqrt{\frac{1 + \frac{1}{2}\mathrm{SR}^2_{\text{period}}}{T_{\text{periods}}}} \times \sqrt{\text{periods per year}} \approx \frac{1}{\sqrt{T_{\text{years}}}}.$$

With five years of data, a strategy with a true annual Sharpe of 0.5 will show a measured Sharpe anywhere between roughly -0.2 and 1.2 (90% range). Fat tails and negative skew widen this further. Figure 1 shows the range narrowing with years of data, and shows twelve strategies with *exactly zero* edge. Several look like careers.

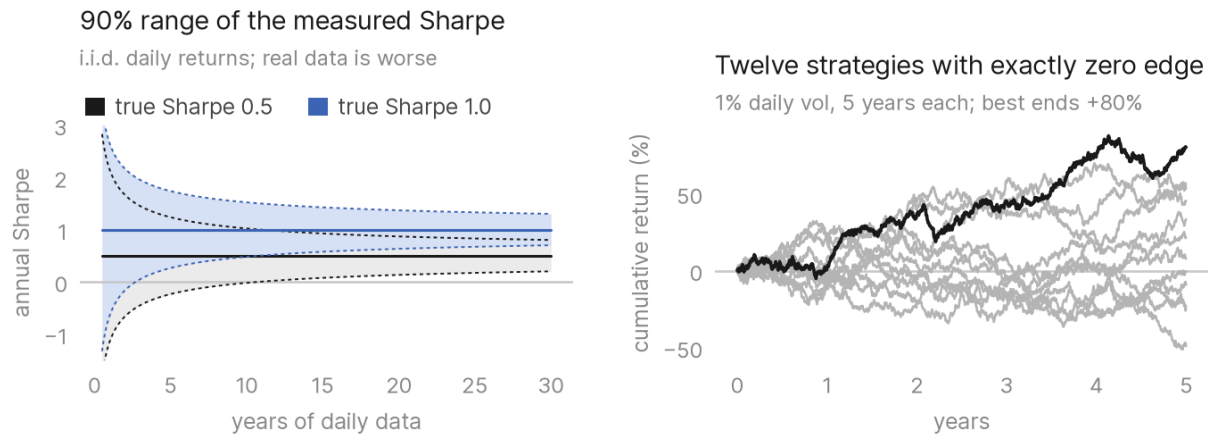


Figure 1. Left: 90% range of the measured annual Sharpe ratio for true Sharpes of 0.5 and 1.0 as data accumulates (i.i.d. daily returns; real data is worse). Even ten years cannot separate 0.5 from 1.0 reliably. Right: twelve random walks with zero expected return. The best of them (black) is up 80% after five years.

2.5 Hypothesis tests and what a p-value is

A test asks how surprising the data would be if there were no effect (the *null hypothesis*). The *p-value* is the probability, under the null, of a result at least as extreme as the one observed. It is *not* the probability that the strategy is worthless. It is not the probability the result will repeat, and it says nothing about whether the effect is large enough to matter. A p-value of 0.03 on an edge smaller than the spread is useless, and a p-value of 0.2 on a large edge measured over too few trades is an open question, not a rejection.

2.6 Base rates: why most significant backtests are false

Bayes' rule links the probability that a passing result is real to three numbers: the share of ideas that are real (the *prior*), the probability a real edge passes (the test's *power*), and the probability a dead idea passes (the significance level α):

$$P(\text{real} \mid \text{pass}) = \frac{\pi \cdot \text{power}}{\pi \cdot \text{power} + (1 - \pi)\,\alpha}.$$

In competitive markets the prior is low. If 2% of tested ideas are real, power is 50% and $\alpha = 5\%$, then of 1,000 ideas 10 real ones pass and 49 dead ones pass: **83% of "significant" results are false** (Figure 2). This is the same argument Ioannidis made about published medical research

[16]. A framework can raise the probability that a pass is real in only three ways. It can raise the prior (better ideas, specific mechanisms), raise power (more data, better measurement), or lower the effective α (fewer, pre-registered tests, and corrections for the ones you did run).

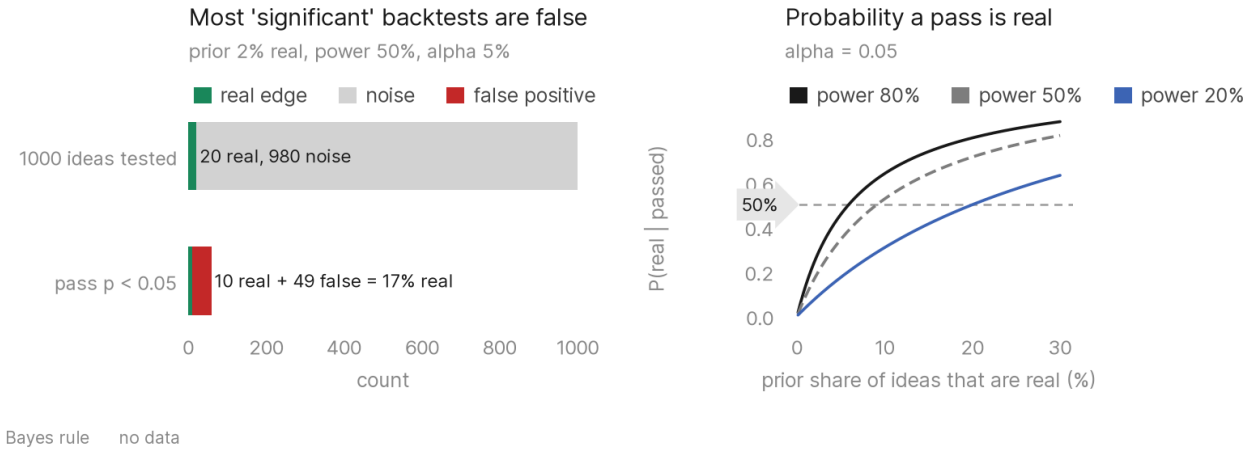


Figure 2. Left: 1,000 tested ideas, 2% real. At power 50% and α = 5%, the passing set holds 10 real and 49 false results. Right: the probability that a pass is real against the prior, for three levels of power. Below a prior of about 5%, most passes are false even with good power.

2.7 Confidence intervals and the minimum useful effect

A 90% *confidence interval* is a range built so that, over many repetitions, it contains the true value 90% of the time. It shows the size of the effect and the uncertainty together, which is why verdicts in this framework are read from intervals. The second ingredient is the **minimum useful effect** (MUE): the smallest net edge worth trading, fixed before the data is seen (for example "at least one round-trip cost per trade, net"). A verdict then comes from comparing the interval with zero and with the MUE, not from asking whether $p < 0.05$ (Section 4.9).

3 How a backtest lies

3.1 Many attempts: the best of noise

Test N strategies with no edge, and the best one's Sharpe ratio grows with N. Bailey and López de Prado [1] give the expected maximum:

$$\mathbb{E}\big[\max_n \widehat{\mathrm{SR}}_n\big] \approx \sqrt{V}\left((1-\gamma)\,\Phi^{-1}\big(1-\tfrac{1}{N}\big) + \gamma\,\Phi^{-1}\big(1-\tfrac{1}{Ne}\big)\right),$$

where V is the variance of the trial Sharpe ratios, $\gamma \approx 0.5772$ and Φ^{-1} is the inverse normal distribution function. With five years of daily data, trying about 50 configurations is enough to expect an annual Sharpe above 1.0 from pure noise (Figure 3). The right panel shows what happens next. The in-sample winner of 200 noise strategies, picked after five years, goes nowhere afterwards.

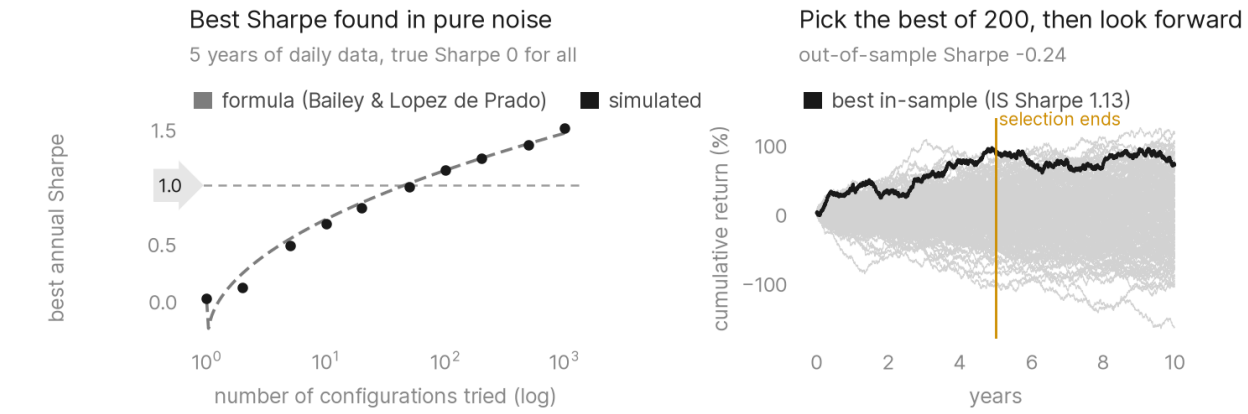


Figure 3. Left: best annual Sharpe among N zero-edge strategies (five years of daily data); dots are simulated, the dashed line is the closed form. Right: 200 zero-edge strategies (grey); the one with the best in-sample Sharpe (black) is chosen at year five; its out-of-sample Sharpe is -0.24.

"Attempts" include more than formal runs. Every parameter tweak, every chart that changed your mind, every abandoned variant and every alternative stop rule is a trial. Statisticians call this the *garden of forking paths*. A framework must count these **hidden trials** too, or its corrections are applied to the wrong number.

3.2 Overfitting: in-sample rank says nothing

Overfitting is what Section 3.1 looks like inside one model. A grid of parameters is fitted to the noise in one sample. Figure 4 runs 78 moving-average crossover rules on 30 independent random walks, picks each walk's in-sample winner, and measures it on the following three years. The winners averaged an in-sample Sharpe of 0.82, and their out-of-sample average was -0.01. Sorted into in-sample deciles, the out-of-sample results are flat. The ranking carried no information.

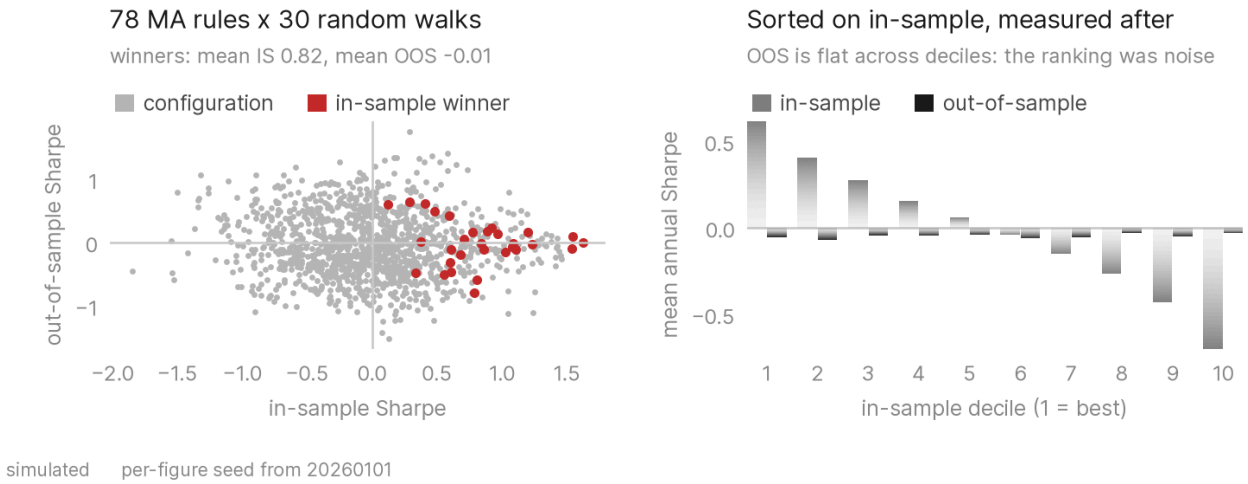


Figure 4. Left: in-sample against out-of-sample Sharpe for moving-average rules on random walks; red points are each walk's in-sample winner. Right: mean Sharpe by in-sample decile. In-sample performance falls steadily by construction, and out-of-sample performance stays flat.

3.3 Leakage: information from the future

Leakage is any route by which information unavailable at decision time enters the decision. It is the most dangerous failure because it produces results that look excellent *and* robust. Figure 5 uses one rule twice on a random walk: trade in the direction of a moving average's slope. The centred average (which uses ten future bars) earns an annual Sharpe of 2.6. The trailing average earns noise. The right panel shows the cheapest general detector, the **truncation test**. Recompute each feature using only data up to time t. A causal feature does not change, and a leaking one does.

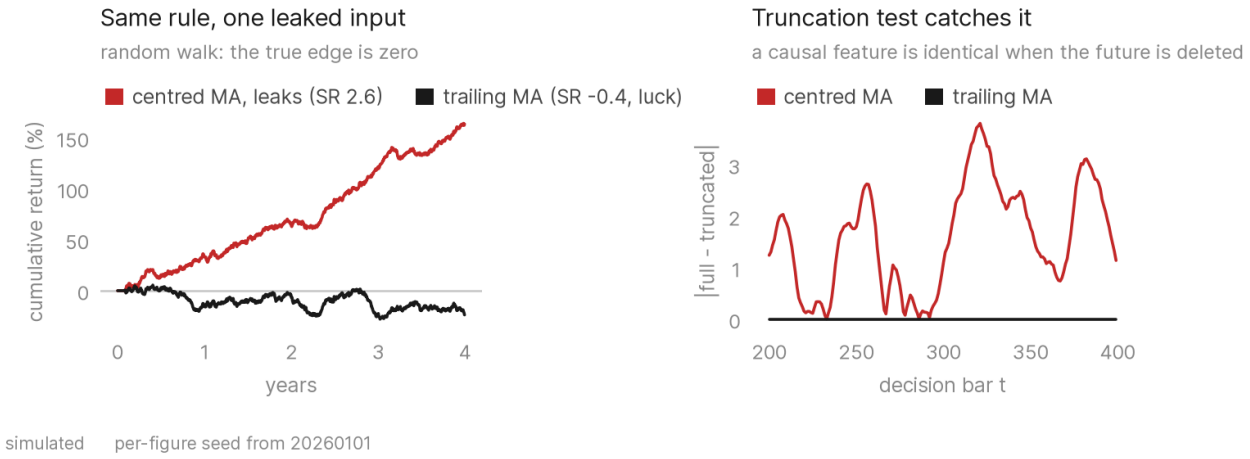


Figure 5. Left: the same slope-following rule using a centred (leaking) and a trailing (causal) moving average on a random walk. Right: the truncation test. The difference between each feature computed on full history and on history cut at t is zero for the causal feature and large for the leaking one.

Table 1. Common leaks and the guard against each.

leak	typical example	guard
look-ahead feature	centred filter, full-sample z-score or scaler	truncation test; fit inside training folds only
overlapping labels	5-day forward return label with adjacent folds	purging and embargo between folds and tiers
publication lag	macro figure used on its reference date	stamp every value with when it became known; as-of joins
revisions	today's revised GDP used in a 2009 decision	vintage (first-release) data
survivorship	today's index members or listed funds; only markets that later became liquid	point-in-time universes
construction	zigzag pivots and swing points, which are only known after confirmation	timestamp at confirmation
execution	fill at the close of the bar that generated the signal	fill at the next bar, crossing the spread
peeking	"just a quick plot" of the test period	tier lock in the data loader; book hidden trials

3.4 Costs and frequency

Costs are certain and edges are not. Suppose each additional trade per day dilutes the edge, so the edge per trade scales like $\sigma/\sqrt{N}$, while each round trip costs c. Then daily gross profit grows like $\sqrt{N}$ and daily cost grows like N. With k the fraction of a daily sigma captured at one trade a day, net profit peaks at $N_{\text{opt}} = N_{\max}/4$, where the break-even frequency is

$$N_{\max} = \left(\frac{k\,\sigma_d}{c}\right)^2 .$$

Measure costs in units of the sigma of the move you trade, not in currency. A contract that looks cheaper per trade can be more expensive in sigma (Figure 6).

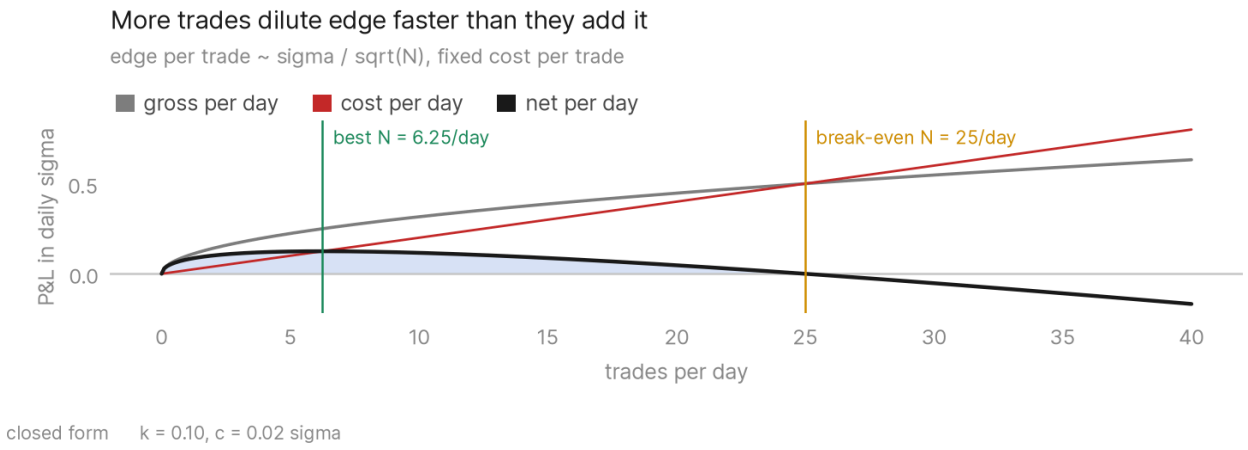


Figure 6. Gross profit grows with the square root of trade frequency, and cost grows linearly. Net profit (black, blue fill) peaks at a quarter of the break-even frequency. The numbers are illustrative.

4 The workflow, from idea to deployment

Figure 7 shows the twelve stages. Two properties make it work. Each stage **writes to a ledger**, an append-only, hash-chained log, so the record cannot be quietly edited. And a failure at any stage is a **recorded verdict**, never a reason to loop back and re-tune. Only three stages read data that the selection process never saw (shaded), and each of them reads it once.

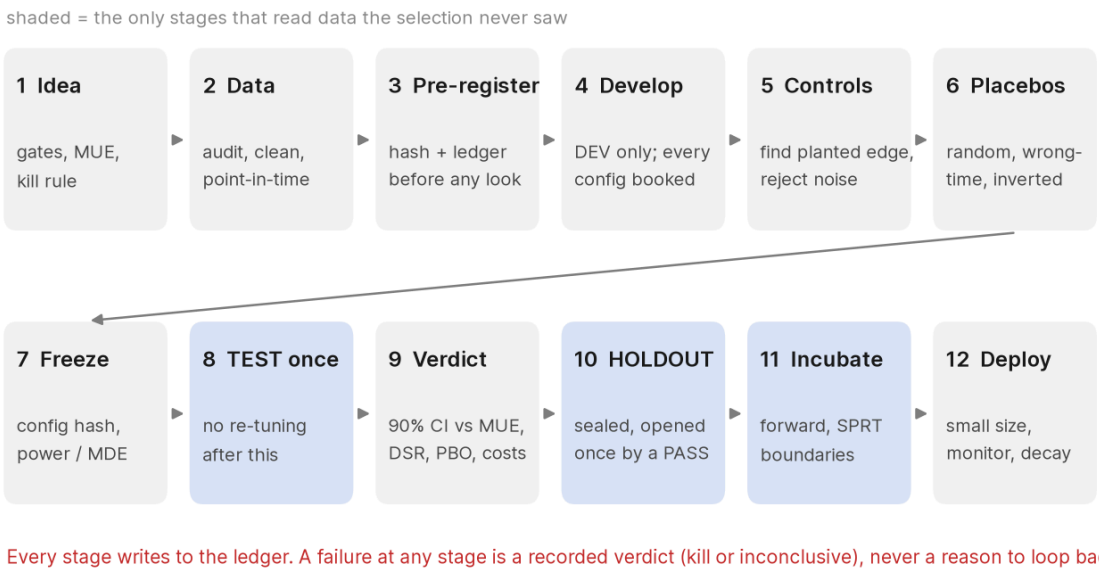


Figure 7. The research workflow. Shaded stages read unseen data. Everything before stage 8 happens on the development tier.

4.1 Idea and gates

Before any code, an idea passes through a few questions. Is there a *mechanism*, someone who must trade regardless of price (index rebalancing, futures rolls, hedging flows, forced liquidation)? A mechanism is optional, but it raises the prior and usually yields a second, independent prediction. Is the effect *observable* in data you actually have? Name the file, the column and the frequency. How many people have already searched for it? Why would it persist? Finally, the **minimum useful effect** and the **kill criterion** are written down. An idea fails at this stage only if the effect is not observable or if no MUE is stated. Everything else is recorded.

4.2 Data

The data stage audits, cleans and documents the data. It is covered in Section 6. The output is a data card per dataset, recording what has already been looked at.

4.3 Pre-registration

The hypothesis, the prediction and its sign, the exact trade rule (entries, exits, sizing, fills, costs), the parameter grid and its size, the primary metric, the statistics, the controls, the MUE and the power plan are written down and *hashed* before evaluation. The hash goes into the ledger. Any later change is either an amendment made before evaluation (with a reason) or a new registration. Pre-registration does not stop you exploring. It stops you from presenting the result of exploration as the result of a test.

4.4 Development on DEV

Data is split by date into three **tiers**, fixed before the first registration: **DEV** for exploration and selection, **TEST** for one pass per frozen configuration, and **HOLDOUT**, sealed for a final check. All selection happens on DEV, using purged and embargoed cross-validation or walk-forward

analysis (Figure 8). A training row whose label window overlaps a validation fold is *purged*, and a gap after each fold is *embargoed*, so the model cannot learn the test labels through overlapping windows [4]. Every configuration evaluated is booked as a trial.

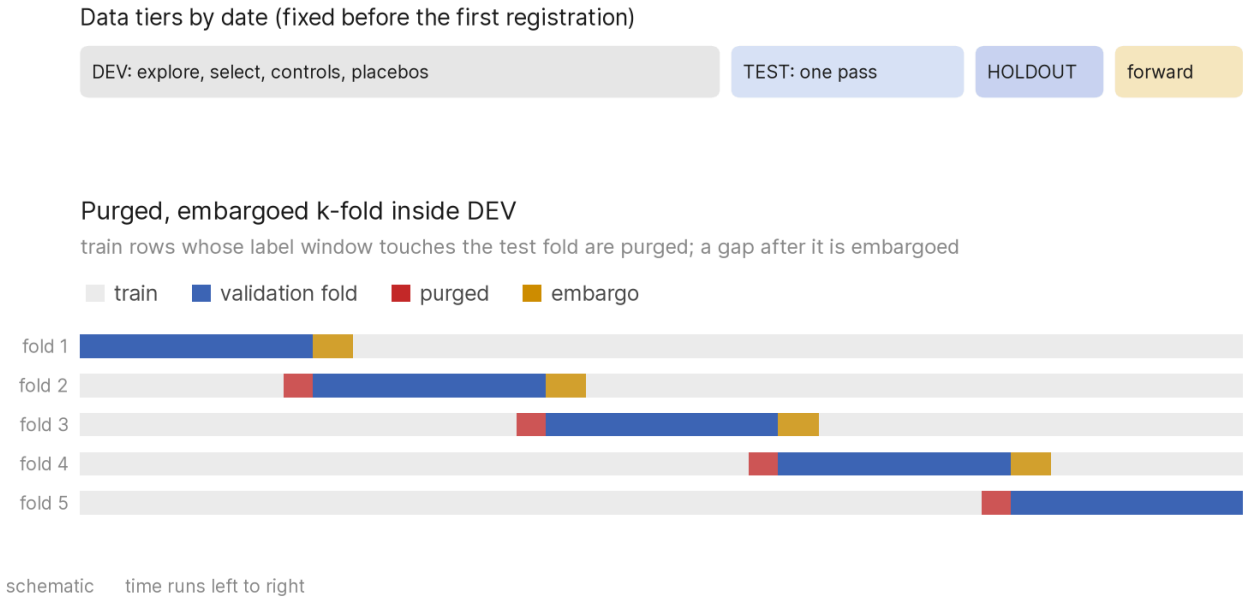


Figure 8. Top: data tiers by date. Bottom: purged, embargoed k-fold cross-validation inside DEV.

4.5 Controls

Before trusting a test harness on a real idea, check that it can see something real and that it ignores noise. A **positive control** plants an edge of known size in the data, and the harness must find it with the right sign and size. A **negative control** feeds the harness noise, and it must find nothing. A harness that fails either test cannot rule anything in or out, and every result it produces is void.

4.6 Placebos

A placebo keeps everything about the strategy except the thing claimed to matter. A trading hypothesis must beat three of them: *random entries* matched on time and count; the rule applied *at the wrong time* (another day or another event, which keeps every marginal distribution intact); and the *inverted signal*. Placebos are judged on gross results, so that costs cannot hide a difference. Figure 9 shows a strategy that clears its placebo distribution and one that does not.

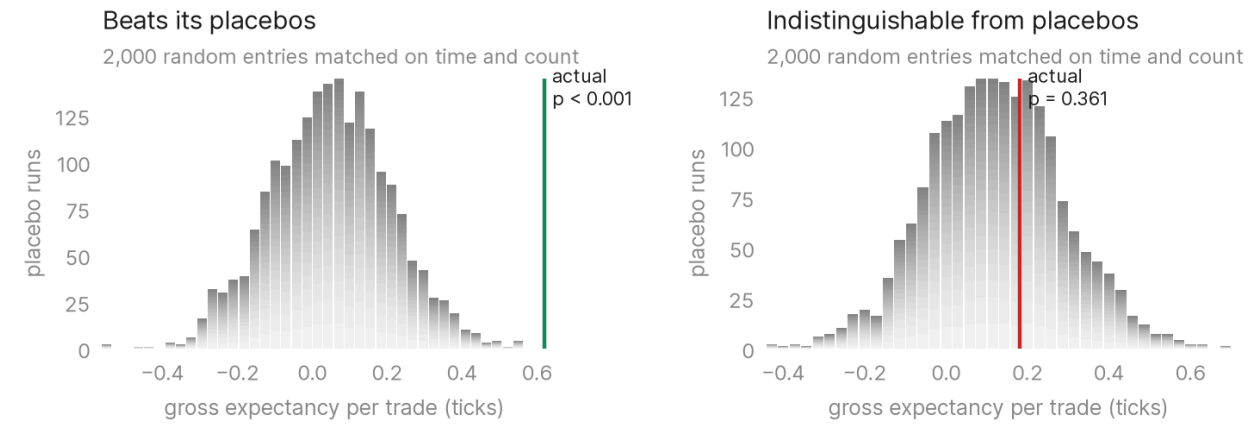


Figure 9. The actual result (vertical line) against 2,000 placebo runs. Left: well outside the placebo distribution. Right: inside it, so whatever the strategy earns, its timing adds nothing.

4.7 Freeze and power

The selected configuration is frozen and hashed. Before TEST is touched, the framework computes the **minimum detectable effect** (MDE): the smallest edge TEST could detect with 80% probability,

$$\mathrm{MDE} = \left(z_{1-\alpha} + z_{0.8}\right) \frac{\sigma}{\sqrt{n_{\text{clusters}}}}.$$

If MDE exceeds MUE, TEST cannot confirm an edge of useful size. The study is registered as underpowered and its best possible verdict is INCONCLUSIVE (Figure 10). Knowing this in advance prevents the most common misreading in research, which is treating "not significant" as "no edge" when the test could never have detected the edge.

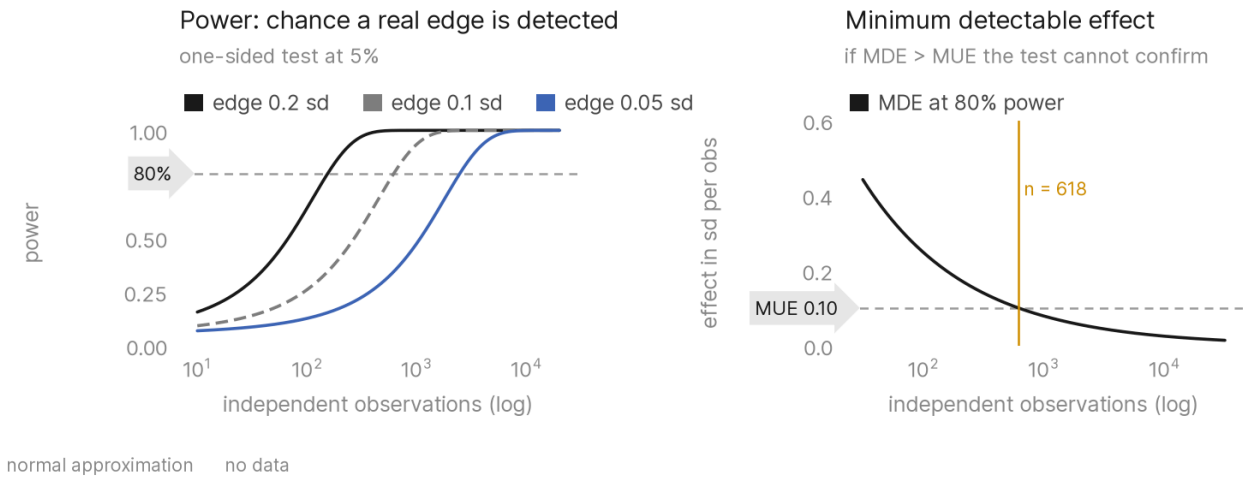


Figure 10. Left: power against sample size for three effect sizes (one-sided, 5%). Right: the MDE falls with the square root of the sample. An MUE of 0.1 standard deviations per observation needs about 620 independent observations.

4.8 TEST, once

The frozen configuration runs on TEST exactly once, and the data loader refuses a second read. If the result disappoints, the finding is negative and is written up as negative. Re-tuning after seeing TEST turns it into more development data, and the next "test" then measures nothing.

4.9 Verdict

The verdict comes from the 90% interval of the primary metric (net of costs) compared with zero and the MUE (Figure 11), together with the gates. **PASS**: the whole interval is above the MUE, every soft check passes (deflated Sharpe, overfitting probability, era agreement), and the stress cost model is still positive. **KILL-CLEAN**: the interval lies wholly below the MUE, or a hard gate failed (leakage, controls, a placebo on a powered sample). **INCONCLUSIVE**: the interval contains both zero and the MUE, or a soft check failed. **KILL-COST**: the edge survives normal costs but not stress costs. Only KILL-CLEAN is terminal. The other classes join a reopen queue, and any reopening is a new registration that adds to the trial count and is tested only on data the original study never touched.

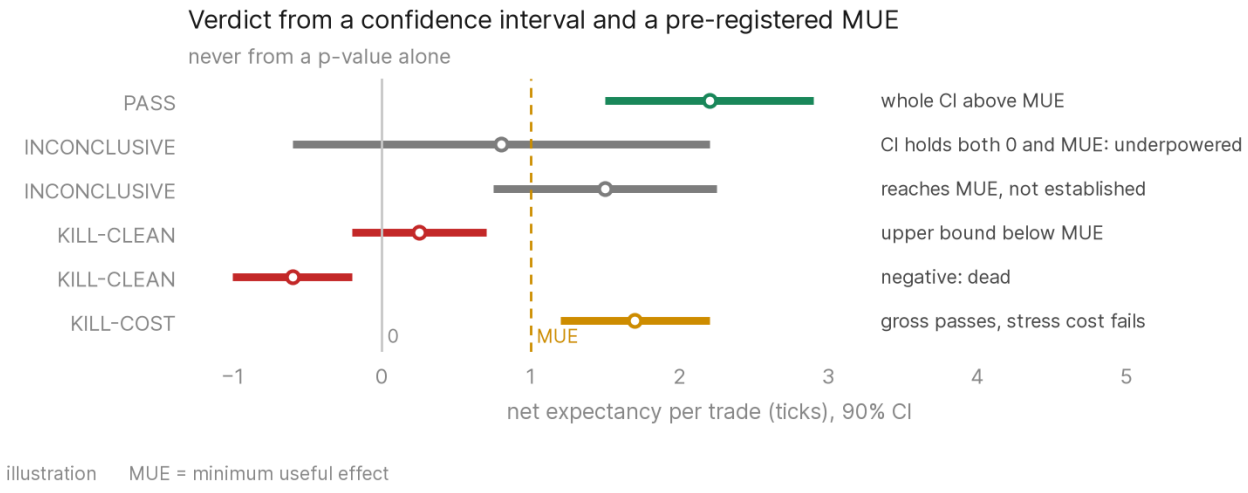


Figure 11. Verdict classes read from a 90% confidence interval against zero and the minimum useful effect.

4.10 HOLDOUT

Only a PASS may open HOLDOUT, once, and only if the configuration hash matches the frozen one, so a re-tuned variant cannot claim it. If HOLDOUT is too short to detect the MUE (MDE > MUE), it can only kill a clear negative. It cannot confirm.

4.11 Forward incubation

The only truly unseen data is data that does not exist yet. The strategy runs on paper or at minimum size, and each period's result is judged by a **sequential probability ratio test** [15] registered in advance. The test accumulates the log-likelihood ratio of "edge as frozen" against "no edge", and stops when it crosses one of two boundaries:

$$\Lambda_n = \sum_{i=1}^{n} \frac{\mu_1 x_i - \mu_1^2/2}{\sigma^2}, \qquad \text{confirm if } \Lambda_n \geq \ln\tfrac{1-\beta}{\alpha}, \quad \text{kill if } \Lambda_n \leq \ln\tfrac{\beta}{1-\alpha}.$$

Sequential testing lets incubation stop as soon as the evidence is decisive, without the inflation of false positives that comes from repeatedly checking a fixed-sample test.

4.12 Deploy and monitor

Deployment starts small. Published anomalies lose about a quarter of their returns out of sample and more than half after publication [17]. Assume decay, monitor realised results against the frozen distribution, and set the conditions for switching off before switching on.

5 The toolkit against luck and overfitting

5.1 Counting trials, and the effective number

The ledger counts every configuration evaluated on any data, hidden ones included. Configurations are correlated (a 20-bar and a 21-bar lookback are almost the same strategy), so corrections use the **effective number of trials**, N_{eff}, computed from the eigenvalues λ_i of the correlation matrix of the candidates' returns. One common estimator is $N_{\text{eff}} = (\sum \sqrt{\lambda_i})^2 / \sum \lambda_i$, which equals 1 for identical streams and N for independent ones.

5.2 The deflated Sharpe ratio

The **probabilistic Sharpe ratio** [3] is the probability that the true Sharpe exceeds a benchmark SR^*, allowing for sample length, skewness γ_3 and kurtosis γ_4:

$$\text{PSR}(\text{SR}^*) = \Phi\left(\frac{(\widehat{\text{SR}} - \text{SR}^*)\sqrt{T-1}}{\sqrt{1 - \gamma_3 \widehat{\text{SR}} + \frac{\gamma_4 - 1}{4}\widehat{\text{SR}}^2}} \right).$$

The **deflated Sharpe ratio** (DSR) sets the benchmark to the Sharpe that the best of N_{eff} noise trials would be expected to reach (Section 3.1) [1]. DSR ≥ 0.95 means the result is unlikely to be the luckiest of the trials that were run. A good p-value with a low DSR means one thing: there was too much searching for the amount of evidence available. More data is needed, not more tests. The gap $\widehat{\text{SR}} - \text{SR}_0$ is the "haircut" Sharpe. Apply only this one deflation, never a second discretionary haircut on top.

5.3 Probability of backtest overfitting

Combinatorially symmetric cross-validation (CSCV) [2] splits the time series into S blocks and considers every way of using half of them for selection and half for evaluation. For each split it takes the in-sample winner and finds its rank out of sample. The **probability of backtest overfitting** (PBO) is the share of splits in which the winner falls below the out-of-sample median. Pure noise gives PBO ≈ 0.5. A grid whose members genuinely differ gives a low PBO (Figure 12). A PBO above about 0.3 means the parameter choice carries too little information to rely on.

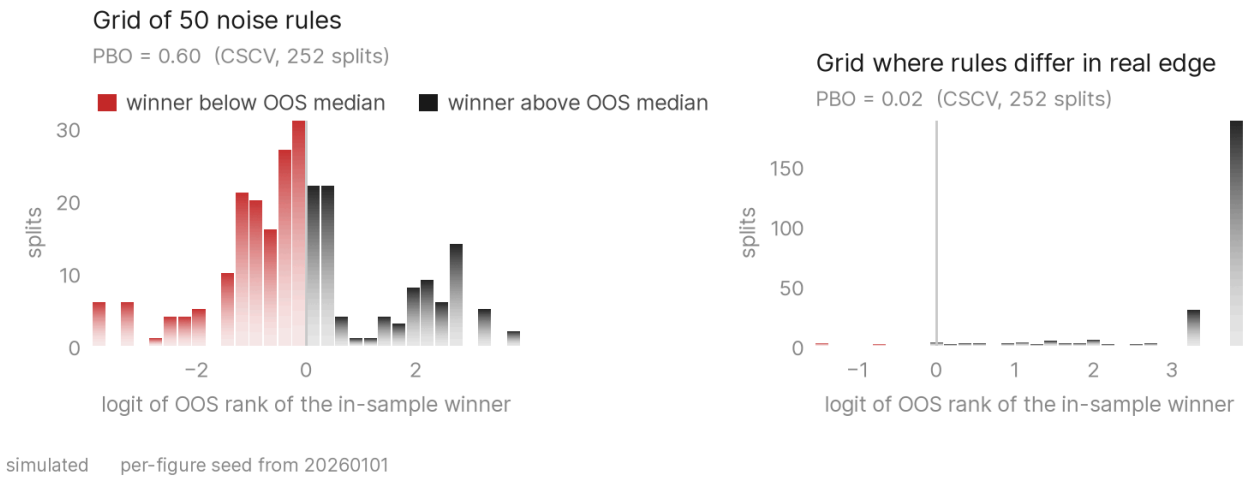


Figure 12. CSCV logits of the out-of-sample rank of the in-sample winner. Left: 50 noise rules, PBO = 0.60 (noise averages 0.5; a single run varies). Right: 50 rules with genuinely different edges, PBO = 0.02.

5.4 False discovery rate for screens

When many candidates are screened at once (a hundred features, fifty markets), the question changes from "is this one real?" to "what share of my discoveries are false?". The Benjamini–Hochberg procedure [9] sorts the m p-values and rejects the k smallest, where k is the largest index with $p_{(k)} \le k\,q/m$. This controls the expected share of false discoveries at q (we use 0.10). Figure 13 shows the difference from a naive 0.05 threshold on 1,000 tests, 100 of them real.

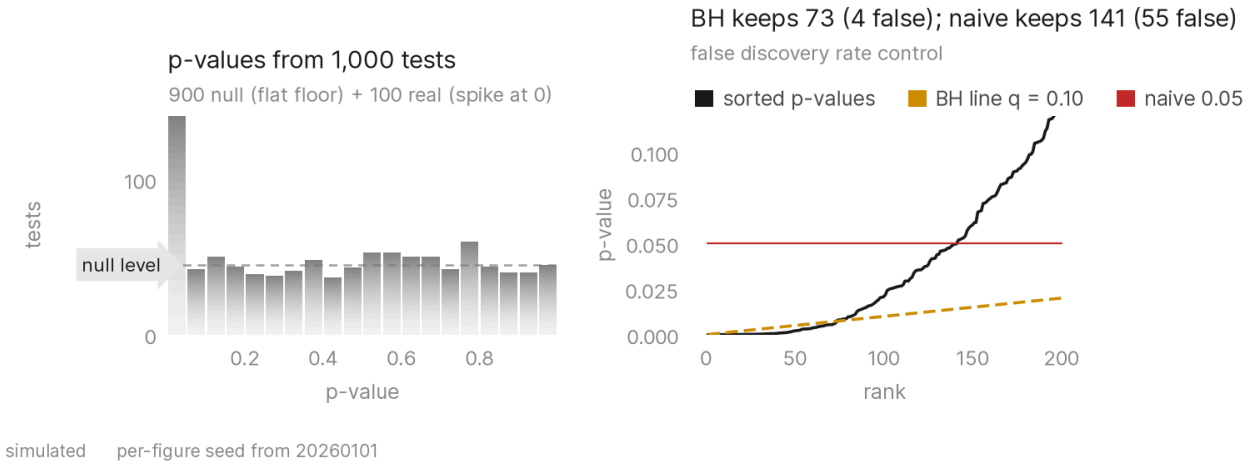


Figure 13. Left: p-values from 1,000 tests. The nulls give a flat floor and the real effects a spike near zero. Right: the BH line keeps 73 discoveries, 4 of them false. A naive 5% cut keeps 141, with 55 false.

5.5 Other checks that cost little

- **Era sign agreement.** Split the data into three or more eras. A real effect has the same sign in each.
- **Parameter neighbourhood.** A good configuration sits on a plateau. A lone peak surrounded by losers is fitted noise.
- **Winner's curse.** Re-run the chosen configuration on a fresh seed or resample before reporting it.

- **Two weightings.** Report the per-cluster series (the sum of each session's trades) and the per-trade series. If their signs or significance disagree, find out why.
- **A higher bar for t-statistics.** Given how many factors have been tried, new claims should clear $t \approx 3$, not 2 [5].

6 Data: aggregation and cleaning

Most research errors are data errors. The pipeline runs in one direction: **raw → clean → derived**. Raw files are never edited. Each one is hashed and recorded in a manifest with its source, retrieval time and licence. Clean and derived data are rebuilt from raw by code, and the rebuild is checked to be bit-identical. The checklist:

1. **Schema and units.** Check columns, types and units against the expected schema on every load.
2. **Time.** Convert timestamps to UTC at ingest. Keep exchange-local time as a derived column. Use the venue's calendar (holidays, early closes, halts). Test the weeks when daylight saving changes, especially when two regions switch on different dates.
3. **Knowledge time.** Give every value an `available_at` stamp, the moment it could have been known, and join datasets *as of* that stamp, never on the nominal date.
4. **Duplicates and gaps.** Drop exact duplicates. Keep distinct fills that share a timestamp, since a single order can produce several. Flag missing and partial sessions against the calendar. Never forward-fill prices across a gap for return calculations.
5. **Outliers and bad values.** Flag rather than delete: prints outside the quote, non-positive prices, crossed quotes, overflowed integers, sentinel values. Log every removal with its reason.
6. **Discontinuities.** Stitch futures rolls (take returns within one contract, or back-adjust), adjust for splits and distributions (total return), and handle index re-bases.
7. **Point in time.** Use universes, constituents, fundamentals and macro values as they were known then: vintages, not revisions.
8. **Reconciliation.** Where two sources overlap, measure their agreement, and choose the primary source before looking at any returns.
9. **A data card and a quality PNG for every dataset.** Record coverage, gaps, outliers, cleaning counts, and which hypotheses have seen which period.

> THE DATA-STATUS RULE
>
> Any period that anyone has studied for this family of ideas is not a holdout, however long ago the study was. Keep a record of what has been seen. Truly unseen data is usually data recorded after you started.

7 Three domains

7.1 Intraday

The cluster is the session, defined in exchange time. The primary series is each session's net P&L: the *sum* of its trades, zero on days with no trade. Fills happen at the next bar and cross the spread. Stops fill at least one tick through, with slippage measured from the data. Any

ambiguity inside a bar goes against the trade. Common traps: features anchored to the session open (running highs, session VWAP) that are really clocks, because their distance from price grows through the day; levels that predict volatility but not direction; and results that come from the illiquid expiring contract around a roll. If the strategy will be traded in an evaluation account with drawdown rules, the account is a barrier problem. Simulate its rules on the intraday path, after checking the simulator against closed forms. For example, at zero drift a static floor at distance D below a target T passes with probability $D/(T+D)$.

7.2 Long-horizon portfolios

The cluster is the rebalance period, so samples are small: twenty years of monthly data is 240 observations, and much less once the observations are made independent. Use total returns, decide on published closes and fill at the next tradable price. Use point-in-time universes, since a rule that chooses from today's list of funds is biased. Compare against a deployable benchmark scaled to the same volatility. Because superiority is hard to establish with so few periods, a **non-inferiority** verdict is useful: the strategy is not worse than the benchmark by more than a pre-set margin, and it is cheaper, simpler or less volatile. Stress the result by changing the start year, by dropping the best and worst year, and by testing on markets that were not used for selection.

7.3 Prediction markets

A contract that pays 1 if an event happens trades at a price that looks like a probability, but only after fees, the spread and the cost of locking up capital until resolution. The unit of independence is the **event**. Outcomes of the same event, and events resolving on the same day or in the same category, are correlated. Leakage is easy here. A market list downloaded today contains survivors, and filtering on final volume selects markets that later became interesting. Last or settlement prices are not prices you could have traded at. Use order-book snapshots taken before the decision, and fill at the bid or ask for the size traded. Skill is measured against the market's own forecast. A model's Brier score must beat the market's, its calibration should be checked with a reliability diagram (Figure 14), and its trades should show *closing-line value* (the price moves toward your side after entry). Resolution disputes, voids and rule changes are a separate risk and get their own bucket. Capacity is limited by the depth of the book. An edge that exists only at small size is a measurement, not a strategy.

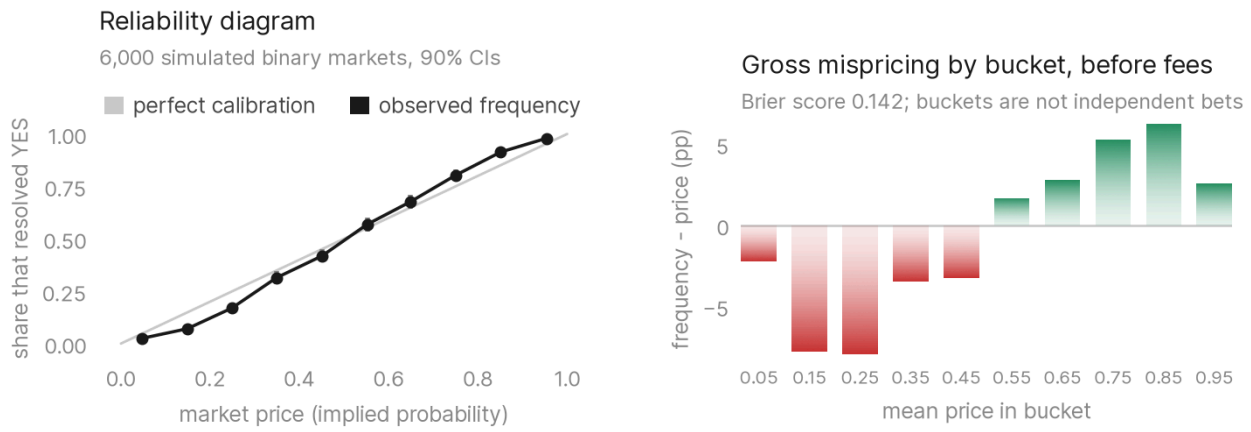


Figure 14. Left: reliability diagram for simulated binary markets with a favourite–longshot bias. Longshots resolve YES less often than their price implies, and favourites more often. Right: gross mispricing by price bucket before fees. Buckets share events and dates, so they are not independent bets, and fees can remove the whole difference.

8 Reading results: a field guide

Every test in the framework produces a verdict sheet (one PNG) and a finding note. Read them in this order.

1. **Verdict and power first.** What class, and could the test have detected the MUE? An INCONCLUSIVE with MDE > MUE means "unknown", not "no".
2. **The interval, not the point.** Where does the 90% interval sit relative to zero and the MUE?
3. **How many tries.** N_{total}, N_{eff}, the DSR and the PBO. A great Sharpe after 300 tries is a small Sharpe.
4. **Placebos and controls.** Did the harness find the planted edge? Did the strategy beat the wrong-time placebo? A failure on the wrong-time placebo means there is no timing information.
5. **Costs.** Look at gross against net, the cost sweep and the break-even cost. "Negative net, positive gross" means the effect is real but smaller than the spread.
6. **Stability.** Era signs, the parameter neighbourhood, and the start-date sensitivity.
7. **The picture.** The equity curve with bootstrap bands, and the drawdown. A curve that is too smooth is a warning.

Table 2. Red flags that warrant an audit before belief.

what you see	usual explanation
annual Sharpe above about 3 on daily data, or a near-straight equity curve	leakage, fills at impossible prices, or a cost model that is missing
edge concentrated in one era or one episode	a regime, not a rule; check era signs and drop the best year
a lone optimum, with neighbours losing	fitted noise (check PBO and the neighbourhood)
good p-value, DSR below 0.95	too much searching for the evidence available
per-trade and per-session results disagree	a few crowded days dominate; weighting artefact
edge vanishes when stop slippage doubles	the profits come from optimistic stop fills
result depends on the exact start date	start-date luck; rerun with every start year
strategy beats random entries but not the wrong-time placebo	it captures the market's drift or volatility, not timing

A worked reading

Suppose a report reads: *opening-drive continuation;* $N_{\text{total}} = 48$*,* $N_{\text{eff}} = 14.3$*; TEST net expectancy 0.9 ticks per trade, 90% CI* $[-0.2, 2.0]$*; MUE 1.5 ticks; MDE 2.1 ticks; DSR 0.62; PBO 0.34; random-entry placebo* $p = 0.03$*, wrong-day placebo* $p = 0.21$*; verdict INCONCLUSIVE (underpowered)*. Read it as follows. The interval contains both zero and the MUE, and the MDE is above the MUE, so TEST could never have confirmed a useful edge. The DSR says that, after 14 effective tries, the Sharpe is unremarkable. A PBO of 0.34 says the parameter choice is weakly informed. The wrong-day placebo performing nearly as well says the entry timing adds little beyond being in the market at that hour. Nothing here is a kill, and nothing here should be traded. The honest next step is more data or a sharper mechanism, registered as a new hypothesis tested on data this study never touched.

9 Working with an AI agent

An agent can build the workspace, write the engines, clean data, run the protocol and draft reports far faster than a person. It also has failure modes that match the problems above. It tends to please, so it may relax a rule that blocks a good-looking result. It explores freely, so it may peek at a test period "to check the data". And it retries freely, which means silent re-runs. The framework answers each of these structurally rather than by trust.

- **Locks, not promises.** The data loader refuses TEST and HOLDOUT outside the protocol. The ledger refuses edits. The registry refuses incomplete specifications.
- **Everything counted.** The agent books hidden trials, including notebook and chart exploration, so the deflation uses the real number of tries.
- **Ambiguity goes to you.** When a rule is unclear, the agent logs a conflict and asks. It never decides in the result's favour.
- **Pictures.** Every test produces a PNG that the agent must look at before reporting. Many construction bugs are visible long before they are statistically detectable.

Your job is to supply ideas, set the MUE, approve amendments and reopenings, and audit. Useful audit questions: "What is the power statement?" "How many configurations were evaluated, including exploration?" "Show me the wrong-time placebo." "Which features passed

the truncation test?" "What would make this result wrong?" And the most useful request of all: *"Here is my idea. Try to kill it."*

10 Conclusion

A research process is judged by the share of what it believes that is true, not by how often it says yes. Noise, searching, leakage and costs create convincing false edges at a steady rate, and none of them shows up in a t-statistic. The defences are simple to state and tedious to enforce: decide before looking, count every look, lock unseen data, compare intervals with a useful effect size, beat placebos, and treat each stage's outcome as final. Enforcing them in code, a ledger and an agent's standing instructions, rather than in good intentions, is what makes a "no" cheap and a "yes" worth something.

Appendix A Formula sheet

quantity	formula / rule
expectancy	$p\,W - (1-p)\,L - \text{cost}$
standard error of a mean	$\sigma/\sqrt{n}$, with n the number of independent clusters
Sharpe ratio (per period)	$\text{SR} = \mu/\sigma$; annualise by $\sqrt{\text{periods per year}}$
Sharpe standard error	$\sqrt{(1 - \gamma_3 \text{SR} + \frac{\gamma_4 - 1}{4}\text{SR}^2)/(T-1)}$
probabilistic Sharpe	$\Phi\big((\widehat{\text{SR}} - \text{SR}^*)\sqrt{T-1}\,/\sqrt{1 - \gamma_3\widehat{\text{SR}} + \frac{\gamma_4-1}{4}\widehat{\text{SR}}^2}\big)$
expected max Sharpe of N nulls	$\sqrt{V}\big((1-\gamma)\Phi^{-1}(1-\frac{1}{N}) + \gamma\Phi^{-1}(1-\frac{1}{Ne})\big)$, $\gamma \approx 0.5772$
deflated Sharpe	$\text{PSR}(\text{SR}_0)$ with $N = N_{\text{eff}}$; require ≥ 0.95
effective trials	$N_{\text{eff}} = (\sum \sqrt{\lambda_i})^2 / \sum \lambda_i$ (eigenvalues of the correlation matrix)
PBO (CSCV)	share of splits where the in-sample winner ranks below the out-of-sample median; noise ≈ 0.5; require ≤ 0.3
Benjamini–Hochberg	reject the k smallest p-values, $k = \max\{i : p_{(i)} \leq i\,q/m\}$
minimum detectable effect	$(z_{1-\alpha} + z_{\text{power}})\,\sigma/\sqrt{n}$; for one-sided 5% and 80% power, $2.49\,\sigma/\sqrt{n}$
P(real | pass)	$\pi\,\text{power}/(\pi\,\text{power} + (1-\pi)\alpha)$
frequency budget	$N_{\max} = (k\sigma_d/c)^2$, $N_{\text{opt}} = N_{\max}/4$
SPRT boundaries	confirm at $\ln\frac{1-\beta}{\alpha}$, kill at $\ln\frac{\beta}{1-\alpha}$
Brier score	$\frac{1}{n}\sum(p_i - y_i)^2$; skill $= 1 - \text{Brier}_{\text{model}}/\text{Brier}_{\text{market}}$
barrier pass, zero drift	static floor: $D/(T+D)$; continuously trailing floor: $e^{-T/D}$

Appendix B Glossary

Cluster
The unit treated as independent: a session, rebalance period or event.

DEV / TEST / HOLDOUT
Date-based data tiers: develop freely, test once, confirm once.

Embargo
A gap left after a validation fold or tier boundary so that information does not leak across it.

Freeze
Hashing the chosen configuration before TEST. Nothing changes after this.

Hidden trial
Any configuration looked at outside a formal run. It still counts.

Ledger
An append-only, hash-chained record of every registration, trial, data access and verdict.

MDE
Minimum detectable effect: the smallest edge the sample detects with 80% probability.

MUE
Minimum useful effect: the smallest net edge worth trading, set before looking.

Negative / positive control
Noise the harness must ignore, and a planted edge it must find.

Placebo
The strategy with its key ingredient removed (random timing, wrong time, inverted sign).

Pre-registration
Writing and hashing the full test plan before evaluation.

Purging
Removing training rows whose label window overlaps the validation data.

Power
: The probability that a test detects an effect of a given size.

Selection family
: The trials whose results led to choosing this hypothesis. This is the N used for deflation.

Truncation test
: Recomputing a feature with the future deleted. A causal feature does not change.

Verdict classes
: PASS, INCONCLUSIVE, KILL-CLEAN, KILL-COST, KILL-ERA (and SUPPORTED / NOT-SUPPORTED for measurements).

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